ProbTSLANet: Probabilistic Time Series Forecasting with Uncertainty Decomposition

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1 Introduction

Deep forecasters [10, 7, 1] attain strong point accuracy, yet their single-valued outputs collapse the predictive distribution and offer decision-makers no calibrated measure of risk. Following [5, 4], total predictive variance decomposes into *aleatoric* uncertainty (irreducible data noise) and *epistemic* uncertainty (reducible model ignorance). Even a calibrated interval is not enough. Practitioners need to know whether a wide band reflects noise or model ignorance, yet most deep forecasters either ignore uncertainty or model one component at a time.

We introduce **ProbTSLANet**, a probabilistic extension of the Time Series Lightweight Adaptive Network (TSLANet) [1] that recovers both components at the cost of a single extra linear head and a test-time loop. Two parallel heads produce a Gaussian predictive distribution $\mathcal{N}(\mu, \sigma^2)$ [8], and MC Dropout [2] collects K stochastic forward passes whose variance across means estimates epistemic uncertainty. Along the backbone axis, sequential forecasters have moved from Transformer attention [10] to patch-based encoders [7] and adaptive spectral filters [1]. Along the uncertainty axis, aleatoric heads [9, 8] and Bayesian or ensemble methods [2, 6] address disjoint components. In contrast, ProbTSLANet recovers decomposed uncertainty from a single model and is evaluated under proper scoring rules [3], which are minimised only by the true predictive distribution. Our contributions are (i) a minimal heteroscedastic head with a stabilising zero-weight and negative-bias initialisation, (ii) a test-time-only MC Dropout decomposition, (iii) a controlled four-step ablation (A1–A4, detailed in Section 3) plus extended variants on the Weather benchmark, and (iv) evidence that calibration on Weather is driven almost entirely by the aleatoric head, with the epistemic share staying below 0.3% and growing only mildly with horizon.

2 Methods

Given a look-back window $\mathbf{X} \in \mathbb{R}^{L \times C}$ of L past timesteps across C variables, the forecaster predicts a mean and variance $(\mu_{t,c}, \sigma^2_{t,c})$ of a Gaussian distribution $\mathcal{N}$ for each of the next H steps, defining the predictive distribution

$$p_\theta(\mathbf{Y} \mid \mathbf{X}) = \prod_{t=1}^{H} \prod_{c=1}^{C} \mathcal{N}\big(y_{t,c};\, \mu_{t,c}(\mathbf{X};\theta),\, \sigma^2_{t,c}(\mathbf{X};\theta)\big)\,. \tag{1}$$

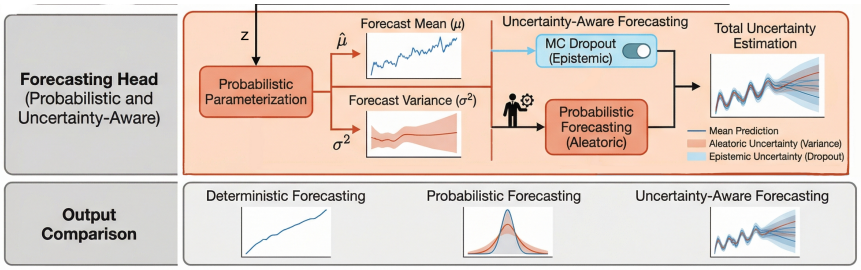

Fig. 1. Probabilistic forecasting head and output comparison. The latent representation $\mathbf{z}$ from the backbone is processed by a probabilistic parameterisation module that produces the forecast mean $\hat{\mu}$ and variance σ^2. The mean path undergoes K=50 MC-Dropout forward passes to estimate epistemic uncertainty, while the learned variance captures aleatoric uncertainty. Both are combined into a total uncertainty estimate. The bottom row contrasts deterministic forecasting (point prediction), probabilistic forecasting (aleatoric bands), and uncertainty-aware forecasting (decomposed aleatoric and epistemic bands).

The shared TSLANet backbone [1] maps the input window to a latent representation $\mathbf{z}$, and Figure 1 shows how the probabilistic forecasting head processes $\mathbf{z}$ into mean and variance predictions with decomposed uncertainty estimates.

Shared backbone. Each input window is centred to zero mean and unit variance, with the inverse transform applied at the output. The normalised window is split into overlapping patches (size 16, stride 8) and embedded to dimension 32. Two stacked TSLANet layers extract features: each applies Layer Norm, an Adaptive Spectral Block (FFT $\rightarrow$ learnable global/local filters $\rightarrow$ energy-adaptive high-frequency mask $\rightarrow$ IFFT), another Layer Norm, and an Interactive Convolutional Block (parallel 1×1 and 1×3 convolutions with multiplicative GELU gating), plus residual connections. Channel independence (processing each variable with shared weights rather than cross-variable attention) keeps the backbone at 32K parameters regardless of variable count. Adding the two output heads brings the total to 100K (H=96) or 269K (H=336).

Heteroscedastic head. Heteroscedastic models predict input-dependent variance rather than a fixed noise level. As shown in Figure 1, the latent representation $\mathbf{z}$ from the backbone is flattened per channel and fed to two parallel linear heads that yield $\mu_{\text{norm}} = W_\mu \mathbf{z} + b_\mu$ and $\log \sigma^2_{\text{norm}} = W_\sigma \mathbf{z} + b_\sigma$, and are denormalised to native scale via $\mu = \sigma_{\mathbf{X}} \mu_{\text{norm}} + \mu_{\mathbf{X}}$ and $\log \sigma^2 = \log \sigma^2_{\text{norm}} + 2 \log \sigma_{\mathbf{X}}$. Two choices stabilise joint optimisation of μ and σ^2. First, W_σ is zero-initialised with $b_\sigma = -2.0$ so the initial predicted variance in the normalised space is a constant $e^{-2} \approx 0.135$, preventing the precision term $1/\sigma^2$ from dominating early gradients. Second, $\log \sigma^2_{\text{norm}}$ is clamped to $[-10, 10]$ for numerical stability.

Training objective. Deterministic baselines use MSE. Probabilistic variants use Gaussian negative log-Likelihood (NLL), dropping the constant $\log 2\pi$,

$$\mathcal{L}_{\text{NLL}} = \frac{1}{2HC} \sum_{t=1}^{H} \sum_{c=1}^{C} \left[\log \sigma_{t,c}^2 + \frac{(y_{t,c} - \mu_{t,c})^2}{\sigma_{t,c}^2} \right]. \tag{2}$$

The first term penalises over-confident variance. The second is a precision-weighted (scaled by $1/\sigma^2$) squared error that down-weights noisy timesteps. Training runs 5 epochs of masked patch pretraining (randomly masking 40% of input patches and training the model to reconstruct them) followed by 15 epochs of supervised fine-tuning on (2), keeping the best checkpoint by validation MSE.

Test-time uncertainty decomposition. We apply Monte Carlo (MC) Dropout [2] at inference. We keep only `nn.Dropout` layers active (not DropPath, which randomly drops entire residual branches and would alter the architecture) and collect K=50 stochastic forward passes $(\mu^{(k)}, \sigma^{2(k)})$. Following [5],

$$\text{Var}[y \mid \mathbf{X}] \approx \underbrace{\frac{1}{K} \sum_{k=1}^{K} \sigma^{2(k)}}_{\text{aleatoric}} + \underbrace{\text{Var}_k \left[\mu^{(k)} \right]}_{\text{epistemic}}, \tag{3}$$

and the final mean forecast is $\bar{\mu} = \frac{1}{K} \sum_k \mu^{(k)}$. Training uses a single forward pass per batch, and only inference pays the K-pass cost.

3 Experiments

We evaluate the contribution of each probabilistic component through a controlled ablation study on a standard forecasting benchmark.

Dataset and protocol. We use the Weather benchmark [10], which has 21 variables at 10-minute resolution, with the standard chronological 70/10/20 train/validation/test split, look-back L=96, and horizons $H \in \{96, 336\}$. We report MSE and MAE for point accuracy. For probabilistic quality we report Gaussian NLL, analytic Gaussian Continuous Ranked Probability Score (CRPS) [3], calibration error (mean $|\text{nominal} - \text{observed}|$ over quantile levels $\{0.1, \ldots, 0.9, 0.95\}$, which a well-calibrated model achieves zero), and mean epistemic and aleatoric variances.

Implementation. The backbone uses 2 TSLANet layers with embed dim 32, patch size 16, stride 8, and dropout 0.3, giving 32K backbone parameters and 100K or 269K total at H=96 or H=336 for the heteroscedastic variants. Training uses AdamW (lr=10^{-4}, weight decay 10^{-6}) with ReduceLROnPlateau (factor 0.1, patience 3) and gradient clipping at max-norm 4.0. We run 5 pretrain epochs followed by 15 fine-tune epochs, draw K=50 MC samples at test time, fix seed 42, and execute CPU-only under `comp0197-pt`.

Table 1. Main results on Weather at horizons H=96 and H=336. Lower is better. Best values within the A1–A4 cascade are bolded.

Model	MSE↓	MAE↓	NLL↓	CRPS↓	Cal. Err.↓
A1-96	**0.1736**	**0.2152**	–	–	–
A2-96	0.1739	0.2154	–	0.2057	0.4363
A3-96	0.1968	0.2229	0.2861	**0.1848**	**0.1023**
A4-96	0.1968	0.2232	**0.1931**	0.1864	0.1115
A5-96	0.1829	0.2172	0.1675	0.1917	0.1306
A6-96	0.1730	0.2148	1.4215	0.4777	0.2562
A7-96	0.1829	0.2174	0.1357	0.1936	0.1386
A1-336	**0.2753**	**0.2959**	–	–	–
A2-336	0.2755	0.2960	–	0.2861	0.4851
A3-336	0.3142	0.3169	**0.6794**	**0.2866**	**0.1572**
A4-336	0.3132	0.3167	0.6811	0.2896	0.1630
A5-336	0.2858	0.3003	0.6337	0.2812	0.1721
A6-336	0.2747	0.2951	3.4762	0.4849	0.2019
A7-336	0.2856	0.3004	0.6355	0.2862	0.1822

Cascading configurations. We design a four-step cascade that isolates each probabilistic component. **A1** is the deterministic MSE baseline with a single forward pass. **A2** reuses A1 weights with MC Dropout at test time and captures the epistemic component only. **A3** trains Gaussian heads with NLL (2) and runs a single deterministic forward pass, giving aleatoric uncertainty only. **A4** combines NLL training with MC-Dropout inference and decomposes the total variance via (3). A1 and A2 share weights, and so do A3 and A4.

Extended ablations. **A5** adds an auxiliary MSE term, $\mathcal{L} = \mathcal{L}_{\text{NLL}} + \lambda \mathcal{L}_{\text{MSE}}$ with λ=0.3, to close the point-accuracy gap. **A6** decouples objectives via two-stage training. Stage 1 trains the backbone and mean head under MSE for 15 epochs, and stage 2 then freezes them and trains only the log-variance head under NLL for 15 further epochs. **A7** combines A5's auxiliary loss with A4's MC-Dropout inference.

4 Results

Point accuracy is largely preserved. Replacing the deterministic head with a heteroscedastic Gaussian head (A1→A3) raises MSE from 0.1736 to 0.1968 at H=96 (+13%) and from 0.2753 to 0.3142 at H=336 (+14%, Table 1). The NLL objective forces the mean head to share capacity with the variance head, and A5 recovers most of this gap (MSE 0.1829 and 0.2858), confirming that the degradation reflects optimisation rather than a fundamental capacity limit.

The aleatoric head drives calibration. MC Dropout alone (A2) leaves point metrics unchanged but is highly miscalibrated (calibration error 0.4363 at H=96, 0.4851 at H=336). A heteroscedastic head (A3) cuts calibration error by 77% (0.1023) at H=96 and 68% (0.1572) at H=336, and lowers CRPS from 0.2057 to 0.1848. Adding MC Dropout on top (A4) leaves MSE and CRPS within 1% of A3 and moves calibration only marginally (from 0.1023 to 0.1115 at H=96,

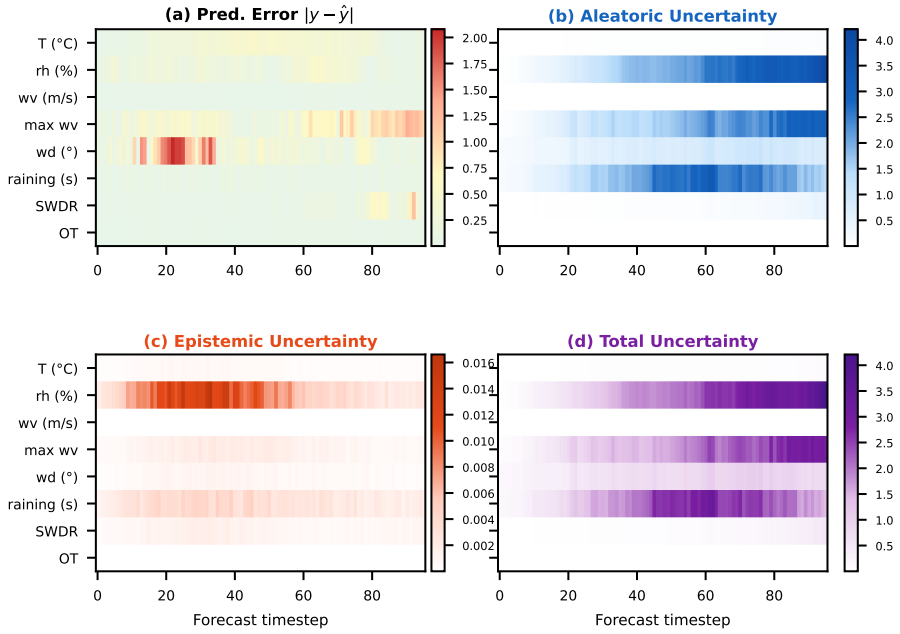

Fig. 2. Uncertainty decomposition heatmap for a test sample (A7, H=96) across eight Weather variables. **(a)** Absolute prediction error. **(b)** Aleatoric (learned) variance dominates, with colorbar range 0–4.0. **(c)** Epistemic (MC Dropout) variance peaks at ∼0.016, confirming the <0.3% share in Table 1. **(d)** Total uncertainty is visually indistinguishable from (b).

and from 0.1572 to 0.1630 at H=336), showing that the aleatoric head does essentially all of the calibration work.

Epistemic variance is a fraction of a percent. Epistemic variance contributes only 0.20% of total predictive variance at H=96 and 0.26% at H=336 in A4, two orders of magnitude below the aleatoric share even though MC Dropout is the canonical epistemic estimator. The fraction grows mildly with horizon but never rivals aleatoric noise, suggesting the lightweight backbone is already near a data-limited regime on Weather. A7 closely tracks A5 on every metric, reinforcing this small-epistemic finding. Figure 2 visualises the decomposition for a representative test sample: the aleatoric panel (b) closely matches the total (d), while the epistemic panel (c) stays near zero (note the colorbar scale difference of ∼250×).

Decoupled training requires decoupled model selection. The extended ablations add a cautionary counterpoint. A6 preserves A1's point accuracy by construction (MSE 0.1730 and 0.2747) but collapses on NLL (1.42 and 3.48, roughly 5× A3). Stage 2 freezes μ yet still selects checkpoints by validation MSE, so log-variance

updates are never weighed against an NLL-aware target and the mean aleatoric variance inflates to 29.4 at H=96 against 0.61 for A3. A6 is therefore best read as a cautionary ablation: decoupling the training objective must extend to the model-selection criterion.

5 Discussion

The cascade delivers a simple takeaway. On Weather, a heteroscedastic Gaussian head is the decisive ingredient for calibrated probabilistic forecasting, while MC Dropout contributes only a small, horizon-dependent epistemic correction. A2 shows that stochasticity without a variance head is miscalibrated. A3 restores calibration at a modest point-accuracy cost. A4 decomposes without substantially reducing the remaining error. A5 recovers point accuracy via auxiliary MSE, while A6 shows that staged decoupling of the training objective must extend to the model-selection criterion, not just the loss.

Limitations. Evaluation is confined to one benchmark, two horizons, and a single seed (42), so the reported gaps are point estimates. MC Dropout approximates an implicit-prior Gaussian posterior that may under-estimate the epistemic share on more complex signals. The small observed epistemic fraction may also reflect the 32K-parameter shared backbone, where data rather than Bayesian disagreement is the binding constraint. The A6 ablation inherits validation MSE as its stage-2 selection criterion, and a fair head-to-head with A3 would require an NLL-based selection rule.

Future directions. Three concrete extensions follow. First, replacing MC Dropout with a last-layer Laplace approximation or SWAG would yield an explicit posterior. Second, extending the Adaptive Spectral Block's energy threshold into a frequency-conditioned variance predictor could route noisy spectral bands directly into the aleatoric head. Third, wrapping the decomposed variance with a conformal post-hoc calibrator would restore finite-sample coverage guarantees when the Gaussian assumption is violated.

Part B: Individual Reflection and Criticism

Personal Contribution. My primary role in this project spanned three areas: literature review, theoretical architecture design, and report writing. I began by studying the TSLANet paper [1] in depth to understand the backbone's Adaptive Spectral Block and Interactive Convolutional Block, as well as its patching, instance normalisation, and channel-independent design. I then surveyed the uncertainty quantification literature, focusing on heteroscedastic regression [8], MC Dropout as approximate Bayesian inference [2], the aleatoric–epistemic decomposition framework [5], deep ensembles [6], and proper scoring rules [3]. This survey informed the theoretical grounding of our extension.

Building on this understanding, I collaborated with Fengyuan Zheng and Flora Xu on the design of the probabilistic extension. Specifically, I contributed to: (i) the choice of dual linear heads that output per-step Gaussian parameters $(\mu, \log \sigma^2)$ rather than a single point estimate, (ii) the stabilising initialisation strategy (zero weights, negative bias) for the log-variance head to prevent early-training dominance of the precision term, (iii) the decision to keep MC Dropout inference-only and to activate only `nn.Dropout` layers (not DropPath), and (iv) the cascading ablation structure (A1–A4) that isolates each probabilistic component. I was not involved in running experiments; my contribution was in the theoretical design and the reasoning behind each configuration. Finally, I was the primary author of the group technical report (Part A), including the mathematical formulation, method descriptions, and result interpretation.

Critical Evaluation. Several specific weaknesses limit the strength of our conclusions. First, all results are based on a single dataset (Weather) and a single random seed (42). Without multi-seed runs or confidence intervals, the reported metric differences—particularly the small gaps between A3 and A4—may not be statistically significant. A paired bootstrap test across multiple seeds would be needed to confirm these findings.

Second, the Gaussian output assumption is violated for several Weather variables. Wind direction is circular and cannot be meaningfully modelled by a unimodal Gaussian; precipitation-related variables are zero-inflated with heavy right tails. While instance normalisation makes residuals more symmetric for well-behaved variables such as temperature, the distributional mismatch likely inflates NLL for the problematic channels. A mixture density network or a distribution-free approach such as conformal prediction would address this.

Third, the epistemic uncertainty estimate from MC Dropout is almost certainly an underestimate. Our backbone has only 32K shared parameters, and the dropout rate of 0.3 applied to a two-layer network provides a limited implicit posterior. Lakshminarayanan et al. [6] showed that deep ensembles consistently outperform MC Dropout in capturing epistemic spread. The $<0.3\%$ epistemic fraction we observe may therefore reflect the weakness of the approximation as much as the true confidence of the model. Replacing MC Dropout with an ensemble of 5–10 independently initialised models, or a last-layer Laplace approximation, would yield a more faithful decomposition.

Fourth, the channel-independent design, while parameter-efficient, discards cross-variable correlations. Weather variables are physically coupled (e.g., temperature and humidity via the Clausius–Clapeyron relation), and modelling their joint distribution could improve both point accuracy and calibration. A lightweight cross-channel attention layer after the backbone could capture these dependencies without sacrificing scalability.

Despite these limitations, the system achieves its core objective. Our deterministic baseline (A1) reaches MSE of 0.1736 at H=96 with only a 2-layer backbone, which is comparable to the results reported by the original TSLANet [1] using significantly deeper architectures. The probabilistic extension preserves this accuracy to within 13% (A3) or 5% (A5, with auxiliary MSE), while pro-

ducing well-calibrated prediction intervals (calibration error 0.10–0.16). If given more time, I would prioritise three improvements: (i) multi-seed evaluation with statistical testing, (ii) replacing MC Dropout with a small ensemble for stronger epistemic estimation, and (iii) extending evaluation to at least one additional dataset (e.g., ETTh1 or Electricity) to test generalisability.

GenAI Audit. I used Claude (Anthropic) in two specific capacities during this project. First, I used Claude as a reading companion during the literature review phase. I adopted a question-and-answer format: after reading a paper (e.g., [5, 2]), I posed specific conceptual questions to Claude to test and consolidate my understanding—for example, asking why the variance of means across MC samples estimates epistemic rather than aleatoric uncertainty, or why NLL is a proper scoring rule for the Gaussian family. I treated Claude's responses as a tutorial aid, not as a primary source; I verified all mathematical claims against the original papers and cross-referenced derivations before incorporating them into our design or report.

Second, I used Claude to polish the language of the group report after I had written the initial draft. Claude's role was limited to improving clarity, grammar, and concision of my existing text; it did not generate new technical content, design choices, or analytical conclusions. All core ideas, mathematical formulations, experimental interpretations, and the first-version writing remained my own work. I reviewed every suggested edit against the technical content to ensure that polishing did not introduce inaccuracies or misrepresent our method. No GenAI tool was used for code generation, experiment design, or data analysis.

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